Adam Shebani

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EDUCATION

University of Wisconsin-Madison, College of Engineering — **Madison, WI**
Degree: B.S. Computer Engineering - Cumulative GPA: 3.9 — September 2025 – December 2027

Madison College — **Madison, WI**
Degree: Pre-Engineering - Cumulative GPA: 3.8 — August 2023 – May 2025

ENGINEERING EXPERIENCE

Silicon Badgers – RTL Team Designer — September 2026 – Present

- Designed a pipelined, parameterizable systolic array in SystemVerilog to accelerate matrix multiplication for the attention and MLP layers of Qwen3.5-2B, using INT4/INT8 MAC processing elements.
- Profiled Qwen3.5-2B inference in llama.cpp, breaking down the compute graph by operation to show matrix multiplication accounts for ~90% of runtime, motivating the systolic array architecture.

ECE 551 – Digital System Design & Synthesis (SystemVerilog) — UW-Madison

- Designed and synthesized digital control logic in SystemVerilog using Synopsys Design Compiler.
- Applied timing constraints and analyzed synthesis reports to meet digital design specifications.
- Integrated an autonomous maze-solving SoC consisting of UART communication, inertial sensor interfaces, PID motor control, and FSM-based navigation in synthesizable SystemVerilog.

ECE 353 – Embedded Systems and Microcontrollers (C) — UW-Madison

- Developed embedded C firmware on an ARM Cortex-M (PSoC 6) using interrupt-driven design with timers, GPIO, PWM, and ADC peripherals.
- Built concurrent I/O functionality using FreeRTOS to coordinate sampling, communication, and display updates without polling.

PROJECTS

Adam16 – 16-bit RISC CPU (SystemVerilog, FPGA/ASIC)

- Designed the RTL for a 16-bit CPU with a custom ISA, FSM-based control unit, and datapath; verified with a 33-case directed testbench achieving full functional coverage.
- Implemented on a Xilinx Artix-7 FPGA (Basys 3) in Vivado, and synthesized to a 32 nm standard-cell library in Synopsys Design Compiler at a 3 ns clock (333 MHz), passing gate-level simulation in Questa.

SFP+ Carrier Board (Altium, 4-layer PCB)

- Designed a 4-layer high-speed PCB with a controlled-impedance stackup: 100 Ω differential and 50 Ω single-ended grounded coplanar waveguide routed to SMA connectors for signal readout.
- Implemented via stitching and a separate chassis ground for signal integrity and EMI, and generated Gerber, BOM, and NC drill outputs for fabrication.

Wearable Motion Tracker (Infineon PSoC 6, C, FreeRTOS)

- Brought up six peripherals over SPI, I2C, and PDM under FreeRTOS on an ARM Cortex-M4: IMU, magnetometer, pressure sensor, 60 GHz radar, MEMS microphone, and QSPI flash.
- Deployed an on-device CNN gesture classifier at 84% accuracy.
- Logged and streamed motion data for offline machine learning workflows.

SKILLS

- C, C++, Python, Java, SystemVerilog
- UART, SPI, I2C, PWM, FreeRTOS
- Synopsys DC, Quartus, ModelSim
- RTL, UVM, STA, FPGA, ASIC
- Git, Linux, Windows
- Oscilloscope, Soldering
- Altium, KiCad, SPICE
- Vivado, Questa, TCL